

A top-down view of a light teal desk. In the top-left corner is a silver laptop with a black keyboard. To its right is a white computer mouse. Above the mouse is a white cup with dark coffee. In the bottom-left corner is a dark brown leather notebook with a white pen resting on it. Next to the notebook are a pair of tortoiseshell-rimmed glasses.

AIClub



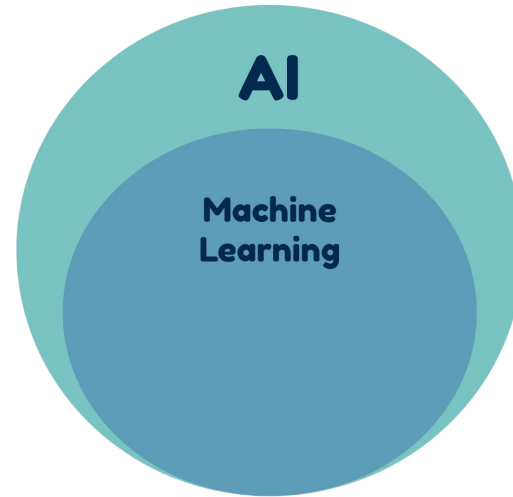


Domains of Artificial Intelligence

Recap – What is AI?

- AI is the ability of machines to perform tasks that require human intelligence. AI can learn, reason, and make decisions.
- Machine Learning is a part of AI.

Quick Question: Give one example of AI.



Why Divide AI into Domains?

- AI performs different types of tasks:
 - Understanding numbers.
 - Seeing images.
 - Understanding language.

To organize these clearly, AI is divided into three domains.



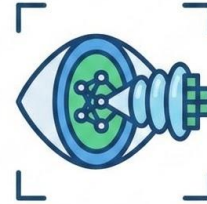
The Three AI Domains

- Data / Statistics
- Computer Vision
- Natural Language Processing (NLP)

Each domain handles a different type of information.



Data / Statistics



Computer Vision



NLP

Domain 1 – Data / Statistics

Definition: AI systems that analyze numbers and data to find patterns and make predictions.

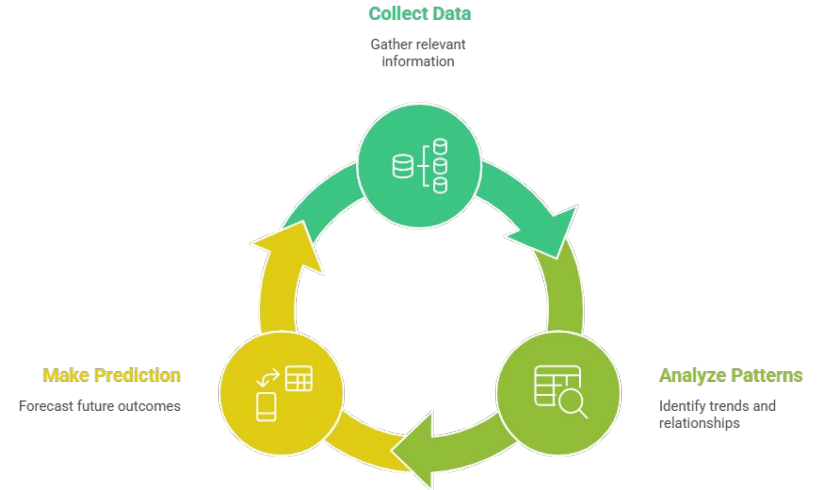
- Works with numbers
- Tables
- Graphs
- Spreadsheets.



How Data Domain Works

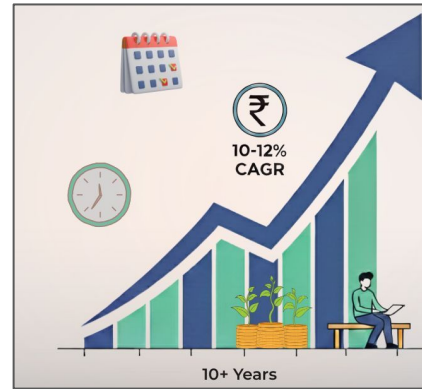
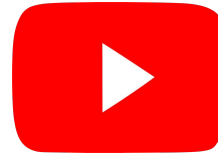
- Collect Data
- Analyze Patterns
- Make Prediction.

Data → Pattern → Prediction



Applications of Data Domain

- YouTube / Netflix recommendations
- Weather forecasting
- Stock market prediction
- Exam result prediction
- Online shopping suggestions
- Fraud detection in banks



Activity – Identify Data Domain

Which of these belong to Data Domain?

- Weather prediction
- Face unlock
- Netflix recommendations
- Chatbot
- Credit card fraud detection.

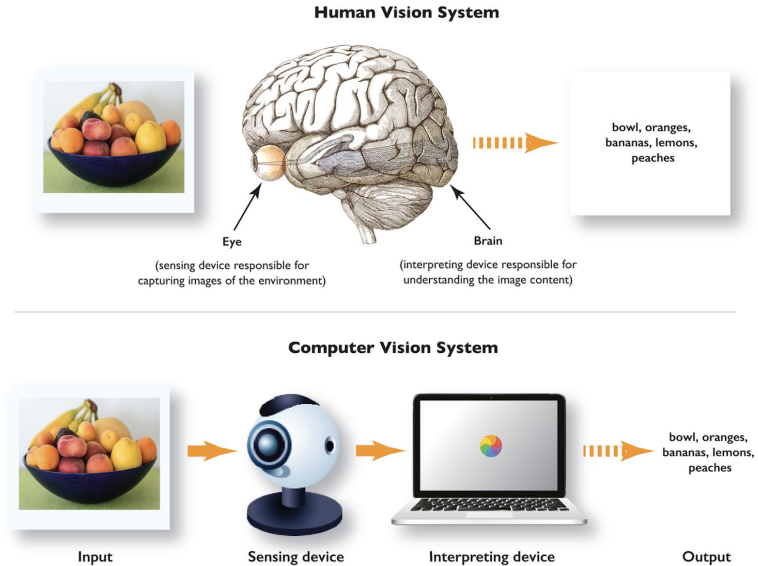


Domain 2 – Computer Vision

Definition: AI systems that can see and understand images and videos.

Works with:

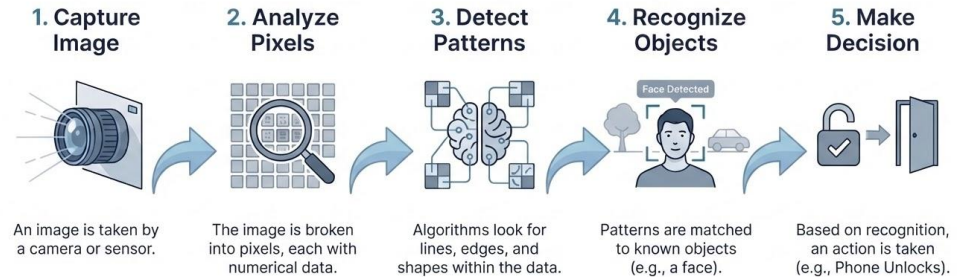
- Photos
- Videos
- Camera input.



How Computer Vision Works

- 1) Capture Image
- 2) Analyze Pixels
- 3) Detect Patterns
- 4) Recognize Objects
- 5) Make Decision.

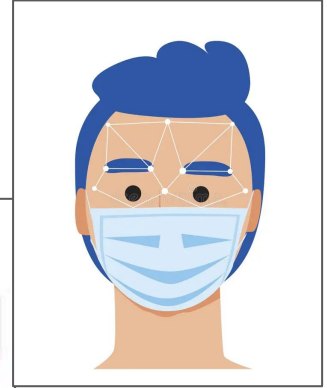
How Computer Vision Works



Example: Face detected → Phone unlocks

Applications of Computer Vision

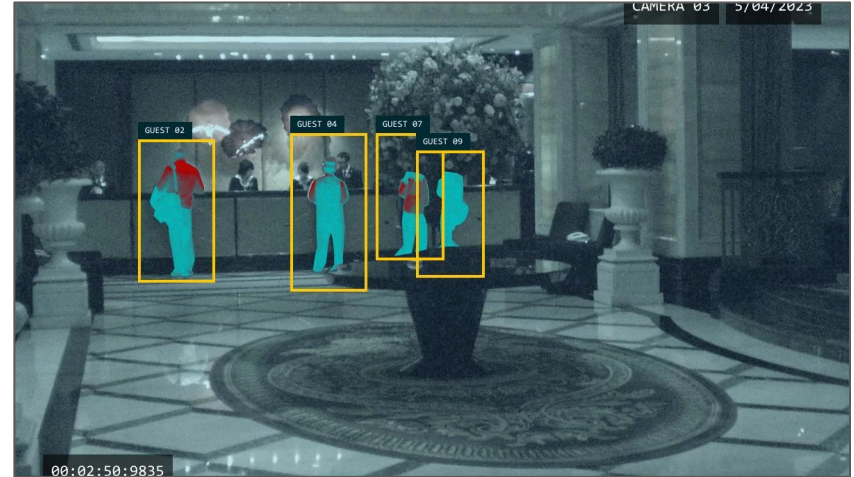
- Face unlock
- Self-driving cars
- QR code scanning
- Medical image detection (X-ray, MRI)
- CCTV surveillance
- Mask detection



Activity – Identify Computer Vision

Which of these belong to Computer Vision?

- Face recognition
- Google Translate
- CCTV surveillance
- Spam email detection
- QR code scanner

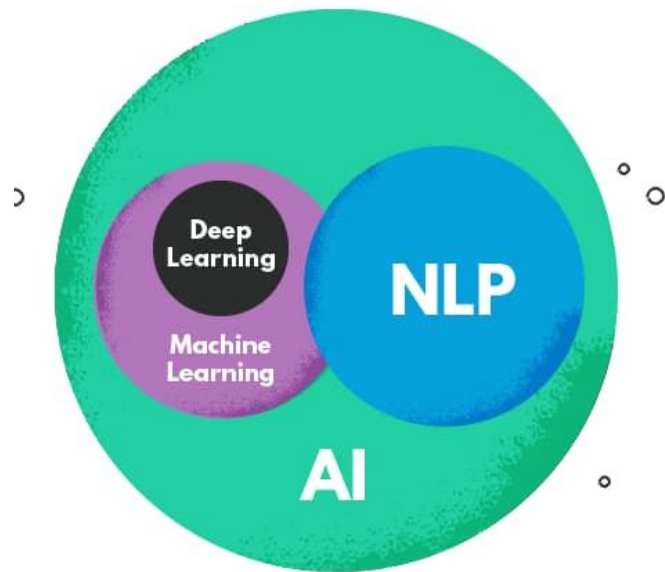


Domain 3 – Natural Language Processing

Definition: AI systems that understand and respond to human language (text or speech).

Works with:

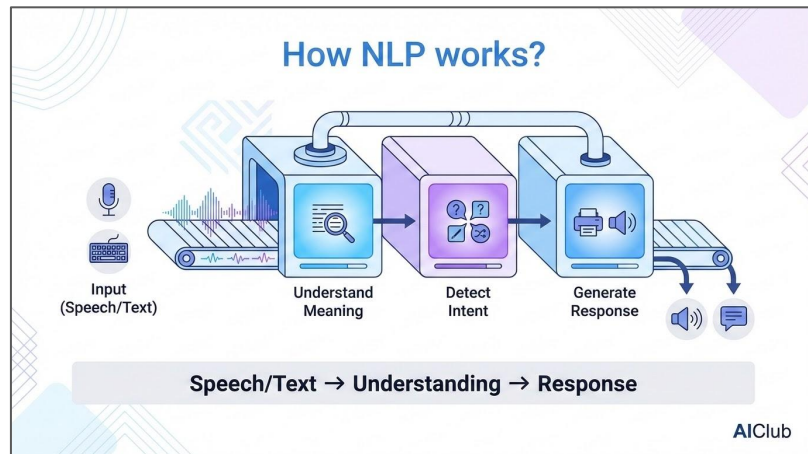
- Words
- Sentences
- Voice
- Text.



How NLP Works

- Input (Speech/Text)
- Understand Meaning
- Detect Intent
- Generate Response

Speech/Text → Understanding → Response



Applications of NLP

- Alexa / Google Assistant
- Chatbots
- Google Translate
- Auto-correct
- Voice typing
- Spam email detection



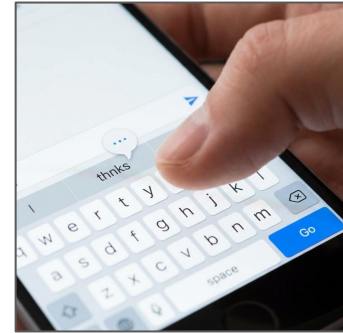
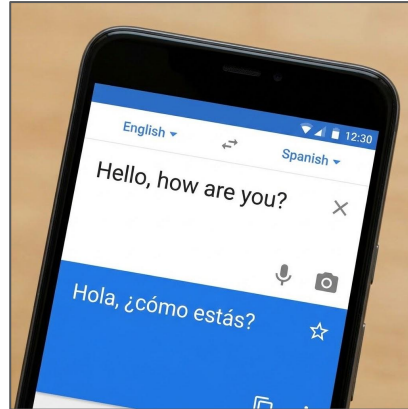
ChatGPT



Activity – Identify NLP

Which of these belong to NLP?

- Chatbot
- Face unlock
- Google Translate
- Weather prediction
- Auto-correct



Comparison Table

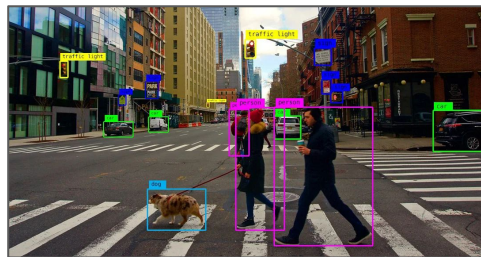
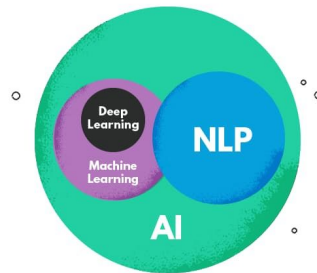
Domain	Work with	Example
Data	Numbers	Weather Prediction
Computer Vision	Images	Face unlock
NLP	Language	Alexa

Can AI Use More Than One Domain?

Example: Self-driving Car

- Computer Vision → Detect objects
- Data Domain → Calculate speed & route
- NLP → Voice commands

One AI system can combine multiple domains.



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THANK YOU



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<https://aiclub.world>